

CM0832 Elements of Set Theory

7. Orderings and Ordinals

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Semester 2017-2

Preliminaries

Convention

The number assigned to chapters, examples, exercises, figures, sections, and theorems on these slides correspond to the numbers assigned in the textbook [Enderton 1977].

Well-Orderings

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Definition

A **structure** is a pair $\langle A, R \rangle$ consisting of a set A and a binary relation R on A .

Transfinite Induction Principle

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Transfinite induction principle

Let $\langle A, < \rangle$ be a **well-ordered** structure and assume that B is a **subset** of A with the special property that for every t in A ,

$$\text{seg } t \subseteq B \quad \text{implies} \quad t \in B.$$

Then B **coincides** with A .

Transfinite Recursion Theorem

Definition

Let $\langle A, < \rangle$ be a **well-ordered** structure and let B a set. The set of **all functions from initial segments of $\langle A, < \rangle$ into B** is defined by

$$B^{A<} := \{ f \mid f : \text{seg } t \rightarrow B, \text{ for some } t \in A \}.$$

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Observation

Let $\langle A, < \rangle$ be a well-ordered structure and let B a set. Note that $B^{A<}$ is a set because we can define it via the subset axiom scheme.

$$B^{A<} := \{ f \in \mathcal{P}(A \times B) \mid f : \text{seg } t \rightarrow B, \text{ for some } t \in A \}.$$

Transfinite Recursion Theorem

Transfinite recursion theorem (preliminary form, p. 175)

Let $\langle A, < \rangle$ be a **well-ordered** structure and let $G : B^{A<} \rightarrow B$. Then there is a unique function F such that for any $t \in A$,

$$\begin{aligned} F &: A \rightarrow B \\ F(t) &= G(F \upharpoonright \text{seg } t). \end{aligned}$$

Replacement Axiom Scheme

Replacement axiom scheme

For any propositional function $\varphi(x, y)$, not containing B , the following is an axiom:

$$\forall A [\forall x (x \in A \rightarrow \exists! y \varphi(x, y)) \rightarrow \exists B \forall y (y \in B \leftrightarrow \exists x (x \in A \wedge \varphi(x, y)))].$$

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Observation

The propositional function φ can depend on other variables $t_1, \dots, t_k$. In this case, we use $\varphi(x, y, t_1, \dots, t_k)$ and we universally quantify on variables $t_1, \dots, t_k$ when using the axiom scheme.

[†]'The membership symbol ($\in$) is not typographically the letter epsilon but originally it was, and the name "epsilon" persists.' [Enderton 1977, p. 182]

Isomorphisms

Definition

Let $\langle A, R \rangle$ and $\langle B, S \rangle$ be two structures. An **isomorphism** from $\langle A, R \rangle$ onto $\langle B, S \rangle$ is a one-to-one function f from A onto B such that for all $x, y \in A$

$$x R y \quad \text{iff} \quad f(x) S f(y).$$

Isomorphisms

Corollary 7H

Let α be the $\in$ -image of a well-ordered structure $\langle A, < \rangle$. Then α is a transitive set and $\in_\alpha$ is a well ordering on α , where

$$\in_\alpha := \{ \langle x, y \rangle \in \alpha \times \alpha \mid x \in y \}.$$

Ordinal Numbers

Idea

To assign a 'number' to each well-ordered structure that measures its 'length'. Two well-ordered structures should receive the same number, if and only if, they are isomorphic.

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Theorem 7I

Two well-ordered structures are isomorphic iff they have the same $\in$ -image.

Ordinal Numbers

Definition

Let $<$ be a well-ordering on A . The **ordinal number** of $\langle A, < \rangle$ is its ϵ -image. An **ordinal number** is a set that is the ordinal number of some well-ordered structure.

Ordinal Numbers

Definition

A set A is **well-ordered by the membership relation** iff the relation

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is a well-ordering on A .

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Definition (other definition of ordinal number)

A set A is an **ordinal number** iff [Hrbacek and Jech (1978) 1999, p. 107]:

- (i) The set is transitive.
- (ii) The set is well-ordered by the membership relation.

Ordinal Numbers

Burali-Forti theorem (p. 194)

There is **no set** to which every ordinal number belongs.

Well-Ordering Theorem

Well-ordering theorem (p. 196)

For any set A , there is a well-ordering on A

Cardinal Numbers

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Observation

Cardinal numbers and initial ordinals are the same numbers.

Rank

Idea

We want to define hierarchy of sets indexed by ordinals:

$$\begin{aligned} V_0 &= \emptyset, \\ V_{\alpha+1} &= \mathcal{P}V_\alpha, \text{ if } \alpha \text{ is a successor ordinal,} \\ V_\lambda &= \bigcup_{\beta < \lambda} V_\beta, \text{ if } \lambda \text{ is a limit ordinal.} \end{aligned}$$

Regularity Axiom

Regularity (foundation) axiom

Every non-empty set A has a member m with $m \cap A = \emptyset$, that is,

$$\forall A [A \neq \emptyset \rightarrow \exists m (m \in A \wedge m \cap A = \emptyset)].$$

References



Enderton, Herbert B. (1977). Elements of Set Theory. Academic Press (cit. on pp. 2, 14).



Hrbacek, Karel and Jech, Thomas [1978] (1999). Introduction to Set Theory. Third Edition, Revised and Expanded. Marcel Dekker (cit. on pp. 20, 21).